

SSH BRUTE FORCE ATTACK — INCIDENT RESPONSE PLAYBOOK

Framework: NIST SP 800-61 Rev.2 | Attack: T1110 Brute Force | Severity: HIGH | Analyst: George Eze | Date: May 12, 2026 | Version: 1.0

NIST PHASE	OBJECTIVE	ACTIONS	TOOLS	EXPECTED OUTCOME
PHASE 1 PREPARATION (NIST §3.1)	Ensure SOC is ready to handle brute force incidents	• Verify Wazuh SIEM is running • Confirm agent ICONIC connected • Validate SSH brute force rules (Rule 5760, 40111) are active • Confirm Active Response enabled	Wazuh Dashboard ossec.conf local_rules.xml	SIEM operational Detection rules active Agent reporting AR configured
PHASE 2 DETECTION & ANALYSIS (NIST §3.2)	Detect, validate and analyse the brute force attack	• Monitor Wazuh for Rule 40111 alert • Confirm source IP and target user • Check attack speed and volume • Validate alert is not false positive • Check attacker IP on AbuseIPDB • Map to MITRE ATT&CK T1110	Wazuh Dashboard auth.log AbuseIPDB VirusTotal MITRE ATT&CK	Attack confirmed real Source IP identified MITRE technique mapped Severity determined
PHASE 3 CONTAINMENT (NIST §3.3)	Stop the attack and prevent further access attempts	• Block attacker IP via iptables • Wazuh Active Response auto-blocks • Verify no active sessions exist • Lock targeted user account if breach is suspected	iptables / ufw Wazuh AR who / w / last usermod -L	Attacker IP blocked No active sessions Endpoint protected Attack stopped
PHASE 4 ERADICATION (NIST §3.3)	Remove attack vectors and harden the system	• Disable SSH password authentication • Set MaxAuthTries to 3 • Disable root SSH login • Install and enable fail2ban • Check for persistence (cron/keys)	sshd_config fail2ban crontab ~/.ssh/ systemctl	SSH hardened Password auth off Root login disabled No persistence found
PHASE 5 RECOVERY (NIST §3.3)	Restore normal operations safely and verify security	• Re-enable user account if locked • Restart SSH with new config • Verify all services running • Monitor dashboard 24-48 hours • Test SSH hardening is working	systemctl Wazuh Dashboard Threat Hunting ssh -p 22 test	Services restored SSH config verified No further attacks System stable
PHASE 6 POST-INCIDENT (NIST §3.4)	Document findings and improve future defences	• Write incident report (1 page) • Update Wazuh detection rules • Present findings to team/manager • Update playbook if needed • Share IOCs with threat intel team	Incident Report local_rules.xml GitHub Email/Slides	Report completed Rules updated Team informed Playbook improved

MITRE ATT&CK MAP T1110 — Brute Force | T1110.001 — Password Guessing | T1021.004 — SSH | T1078 — Valid Accounts | T1595 — Active Scanning

REFERENCES NIST SP 800-61 Rev.2: nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf | MITRE ATT&CK: attack.mitre.org/techniques/T1110